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EX NO: 9

SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF
ENCODED OUTPUT BIT SEQUENCES USING MATLAB

Objectives

- (i) To implement convolutional encoding for a given binary input sequence using a rate convolutional encoder and tabulate the corresponding encoded output pairs.
- (ii) To visualize and compare the input bit sequence and encoded output bit sequence using MATLAB plots to analyze the redundancy introduced by convolutional coding.

MATLAB code

```
clc;
clear;
close all;

% Fixed input sequence
input_bits = [1 0 0 1 1];

% Convolutional encoder
trellis = poly2trellis(3,[7 5]);
encoded_bits = convenc(input_bits,trellis);

% Separate encoded bits into pairs
encoded_pairs = reshape(encoded_bits,2,[]);

% Create output strings
for i = 1:length(input_bits)
    output_pair{i} = sprintf('%d%d', ...
        encoded_pairs(i,1),encoded_pairs(i,2));
end

% Create table
```

```
Bit_Position = (1:length(input_bits));
Input_Bit = input_bits';
Output_Encoded = output_pair';
T = table(Bit_Position,Input_Bit,Output_Encoded);
disp('Convolutional Encoding Table');
disp(T);
% Plot
figure;
subplot(2,1,1)
stairs(input_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Input Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
subplot(2,1,2)
stairs(encoded_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Encoded Output Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
```

